CINEFUND: BLOCKCHAIN-BASED CROWDFUNDING ECOSYSTEM FOR FILM INVESTMEN

A MINI PROJECT REPORT FOR THE COURSE

GE 23627 DESIGN THINKING AND INNOVATION

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BONAFIDE CERTIFICATE

Certified that the mini project titled **CINEFUND: BLOCKCHAIN-BASED CROWD FUNDING ECOSYSTEM FOR FILM INVESTMENT** is the bonafide work carried out by **SUHIRTHA M P (231801175)**, **RAGUL G (231801132)** and **SARATH KUMAR P (231801156)** under my supervision. This project has been completed in partial fulfilment of the requirements of the course **GE23621 – Design Thinking and Innovation**, promoting project-based learning and practical application of design principles. Certified further that, to the best of my knowledge and belief, the work reported herein is original and does not form part of any other project, report, or work submitted previously for any academic award.

This project contributes toward achieving the following **Sustainable Development Goals (SDGs):**

1. SDG 8 - Decent Work & Economic Growth
2. SDG 9 - Industry, Innovation & Infrastructure
3. SDG 10 - Reduced Inequalities

Student Signature with Name

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Signature of the Supervisor with date

Signature Examiner-1 Signature Examiner-

DECLARATION

We hereby declare that the project report titled **"CINEFUND: Blockchain-Based Crowd Funding Ecosystem for Film Investment"** is a bonafide and original work carried out by us under the supervision of [Supervisor Name], [Designation], Department of Artificial Intelligence and Data Science, Rajalakshmi Engineering College, Thandalam, Chennai. This work has been completed in partial fulfilment of the requirements of the course GE23621 – Design Thinking and Innovation. The information presented in this report is true and correct to the best of our knowledge.

Place: Chennai

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ABSTRACT

The Indian film industry, one of the largest in the world, faces a persistent challenge: talented filmmakers and independent producers are unable to bring their creative visions to life due to the dominance of large studios and a lack of accessible funding mechanisms. CINEFUND is a blockchain-based crowd funding ecosystem designed to democratize film investment by enabling direct participation from the public in movie financing.

The platform leverages smart contracts on the Ethereum blockchain to ensure transparent, milestone-based fund release, eliminating intermediaries and reducing fraud. Users can invest amounts starting from ₹500 and earn proportional returns from box office collections, OTT licensing, satellite rights, and overseas distributions once the film generates revenue.

CINEFUND addresses three core problems: inaccessibility of film funding for independent creators, lack of transparency in fund utilization, and exclusion of retail investors from high-value entertainment assets. The system incorporates an escrow mechanism, third-party revenue auditing, KYC-based user verification, and a producer accountability framework.

The platform is built using React.js for the frontend, Node.js and Express for the backend, PostgreSQL as the primary database, and Solidity smart contracts deployed on Ethereum. Key features include a live funding progress dashboard, milestone tracker, profit distribution engine, AI-based movie success scoring, and a secondary investment marketplace.

This report documents the complete design thinking journey — from empathizing with independent filmmakers and retail investors in India, defining the core problem statement, ideating solutions, building and evaluating prototypes, collecting user feedback, and implementing the redesigned system. The project aligns with SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), and SDG 10 (Reduced Inequalities).

CINEFUND represents a transformative step toward an open, transparent, and community-driven film financing ecosystem for India and beyond.

CHAPTER 1

INTRODUCTION

The global film industry generates hundreds of billions of dollars annually, yet the mechanisms for funding film productions remain largely opaque and exclusionary. In India, where cinema is deeply embedded in cultural identity — with over 2,000 films produced annually across multiple languages — the financing landscape is dominated by a small group of studios, production houses, and high-net-worth individuals. Talented independent filmmakers frequently find themselves without access to adequate capital, while millions of cinema enthusiasts have no financial pathway to participate in the success of the films they love.

CINEFUND is conceived as a solution to this structural inequity. By harnessing blockchain technology, smart contracts, and a mobile-first digital platform, CINEFUND enables any individual to invest in a film at any scale — from ₹500 to ₹1,00,000 or more — and receive proportional returns from the film's revenue streams. The platform functions as a regulated, transparent, and community-governed crowd funding ecosystem specifically designed for the entertainment sector.

The system is designed with multiple stakeholder groups in mind: independent producers who need capital, retail investors who seek new asset classes, film fans who want deeper engagement with cinema, and production houses looking to diversify their funding base. CINEFUND provides each of these groups with a dedicated interface, tailored tools, and appropriate transparency mechanisms.

This report documents the complete project development lifecycle using the Design Thinking methodology, moving through stages of Empathize, Define, Ideate, Prototype, Test, and Redesign to arrive at a production-ready solution.

1.1 DESIGN THINKING APPROACH

What is Design Thinking?

Design Thinking is a human-centered, iterative problem-solving methodology that prioritizes understanding the needs of end users before developing solutions. It is widely used in product design, software development, education, and social innovation. Design Thinking moves away from assumption-based development and instead grounds every decision in real-world user research and empathy.

The core principle of Design Thinking is that the best solutions emerge when designers deeply understand the people they are designing for. This methodology encourages teams to challenge assumptions, reframe problems, and create innovative solutions that are both technically feasible and genuinely useful.

Types of Design Thinking Models

Several models of Design Thinking have been proposed by institutions and researchers worldwide:

- Stanford d.school Model (5-Phase): Empathize → Define → Ideate → Prototype → Test — the most widely adopted academic framework.
- IDEO Model: Inspiration → Ideation → Implementation — used extensively in commercial product design.
- IBM Enterprise Design Thinking: Observe → Reflect → Make — adapted for large-scale enterprise software development.
- Double Diamond Model (UK Design Council): Discover → Define → Develop → Deliver — emphasizes divergent and convergent thinking at each stage.
- Google Design Sprint: A 5-day rapid prototyping framework used for startup product validation.

For this project, we adopt the Stanford d.school 5-phase model as our primary framework, as it provides a clear, iterative, and academically structured approach to human-centered design.

1.2 STANFORD DESIGN THINKING MODULE AND ITS PHASES

The Stanford Design Thinking model consists of five non-linear phases that guide teams from problem discovery to validated solutions:

- Phase 1 — Empathize: Conduct user research, interviews, observations, and surveys to understand the experiences, frustrations, and aspirations of stakeholders. For CINEFUND, this involved engaging with independent filmmakers, retail investors, and film enthusiasts across Chennai and Coimbatore.
- Phase 2 — Define: Synthesize research insights into a clear Problem Statement. Use tools like Empathy Maps, Personas, POV (Point of View) statements, and HMW (How Might We) questions to sharpen the focus.
- Phase 3 — Ideate: Generate a broad range of potential solutions through brainstorming, mind mapping, SCAMPER, and other creative techniques. No idea is rejected in this phase — volume and diversity are encouraged.
- Phase 4 — Prototype: Build low-fidelity and high-fidelity representations of the most promising solutions. For CINEFUND, this progressed from paper wireframes to a fully functional web application.
- Phase 5 — Test: Validate the prototype with real users, collect structured feedback, and identify areas for improvement. Testing is iterative — insights from testing feed back into the Empathize and Define phases.

The Stanford model is particularly well-suited to CINEFUND because the problem domain — financial inclusion in the entertainment industry — involves multiple user groups with conflicting needs, requiring deep empathy and iterative refinement.

CHAPTER 2

LITERATURE REVIEW

Domain Papers

[1] Mollick, E. (2014). The dynamics of crowdfunding: An exploratory study. Journal of Business Venturing, 29(1), 1–16. This foundational paper examines the structure of successful crowdfunding campaigns across Kickstarter, identifying the role of social capital, project quality signals, and backer community dynamics. The study established that projects with clear milestones and transparent updates significantly outperform those without, directly informing CINEFUND's milestone-based fund release model.

[2] Belleflamme, P., Lambert, T., & Schwienbacher, A. (2014). Crowdfunding: Tapping the right crowd. Journal of Business Venturing, 29(5), 585–609. This paper differentiates between reward-based and equity-based crowdfunding models. The authors find that equity models are more suitable for capital-intensive projects — such as film production — while reward models work better for creative but lower-budget endeavors. CINEFUND adopts a hybrid model inspired by this distinction.

[3] Zheng, H., Li, D., Wu, J., & Xu, Y. (2014). The role of multidimensional social capital in crowdfunding. Decision Support Systems, 65, 40–50. This paper highlights how social trust networks significantly increase funding success rates. The finding supports CINEFUND's producer rating and review system, which builds trust through verified community feedback.

[4] Agrawal, A., Catalini, C., & Goldfarb, A. (2015). Crowdfunding: Geography, social networks, and the timing of investment decisions. Journal of Economics & Management Strategy, 24(2), 253–274. The authors explore how geographic proximity influences investment decisions in crowdfunding. For CINEFUND's India-first strategy, this insight informs regional language film listing prioritization.

[5] Buterin, V. (2014). A Next-Generation Smart Contract and Decentralized Application Platform (Ethereum Whitepaper). This seminal technical document introduces Ethereum's smart contract

framework, forming the technical backbone of CINEFUND's escrow and profit-distribution mechanism.

Design Thinking Papers

[6] Brown, T. (2008). Design Thinking. Harvard Business Review, 86(6), 84–92. Tim Brown's influential article establishes Design Thinking as a management strategy, arguing that integrating user empathy into business processes leads to more innovative and durable solutions. This paper guided our overall project approach.

[7] Liedtka, J. (2015). Perspective: Linking Design Thinking with Innovation Outcomes through Cognitive Bias Reduction. Journal of Product Innovation Management, 32(6), 925–938. This paper establishes the empirical link between Design Thinking practice and successful innovation, validating the use of this methodology for CINEFUND's development.

[8] Plattner, H., Meinel, C., & Leifer, L. (Eds.) (2011). Design Thinking: Understand – Improve – Apply. Springer. This comprehensive textbook, authored by pioneers of the Stanford d.school model, provided the methodological grounding for our five-phase project structure.

CHAPTER 3

DOMAIN AREA

CINEFUND operates at the intersection of three major domains: Financial Technology (FinTech), Entertainment Technology (EntTech), and Blockchain Technology. Understanding each of these domains is critical to appreciating the scope and complexity of the platform.

3.1 Film Financing in India

The Indian film industry produced over 1,800 certified films in 2023 across Hindi, Tamil, Telugu, Malayalam, Kannada, and other regional languages. The total box office revenue exceeded ₹11,000 crore in 2023, recovering strongly from the pandemic-era lows. Despite this scale, the financing ecosystem remains primitive and exclusionary.

Traditional film financing in India operates through: (a) Studio co-productions where major houses like Eros Now, T-Series, and Lyca Productions fund and distribute; (b) Angel investors — high-net-worth individuals who take equity stakes; (c) Bank loans against distribution guarantees; and (d) Self-financing by producers. Each of these channels excludes the retail public and lacks transparency in fund utilization.

3.2 Blockchain Technology

Blockchain is a distributed ledger technology that records transactions across a network of computers in a way that is immutable, transparent, and decentralized. Each block in the chain contains a cryptographic hash of the previous block, a timestamp, and transaction data, ensuring that historical records cannot be altered.

Smart contracts — self-executing programs stored on the blockchain — allow CINEFUND to automate fund release based on predefined milestones, distribute profits proportionally without manual intervention, and maintain an auditable record of all financial transactions.

3.3 Crowd Funding Platforms

Global crowd funding platforms such as Kickstarter, Indiegogo, and Patreon have demonstrated the viability of public participation in creative project financing. In India, platforms like Ketto and Milaap have succeeded in cause-based funding. However, no platform specifically addresses equity-linked film funding in the Indian context. CINEFUND fills this gap.

3.4 Regulatory Landscape

Film investment in India is regulated by SEBI under the Alternative Investment Fund (AIF) framework. CINEFUND's model is structured to comply with Category I AIF regulations, using a fund manager architecture with proper KYC, risk disclosures, and investor accreditation processes where required.

Domain	Relevance to CINEFUND	Key Technologies
Blockchain	Escrow, smart contracts, transparency	Ethereum, Solidity
FinTech	Payments, KYC, returns distribution	Razorpay, JWT Auth
EntTech	Film data, OTT analytics, revenue tracking	TMDB API, Custom Analytics
Design Thinking	User-centered development process	Stanford d.school Model

CHAPTER 4

EMPATHIZE STAGE

The Empathize stage is the foundation of the Design Thinking process. In this phase, our team conducted primary and secondary research to deeply understand the needs, pain points, motivations, and experiences of our target user groups.

4.1 Research Activities Conducted

- Stakeholder Interviews: 12 in-depth interviews with independent Tamil and Hindi film producers in Chennai and Mumbai.
- User Surveys: Online survey with 120 respondents across age groups 22–45, covering film enthusiast investors.
- Observational Research: Attended two independent film pitch events in Chennai to observe producer-investor dynamics.
- Competitive Analysis: Studied global platforms including Slated, Seed&Spark, and Ketto for film funding.
- Expert Consultation: Spoke with a SEBI-registered investment advisor and a blockchain developer.

4.2 Key User Personas

Persona 1 — Arjun (Independent Film Producer)

Age: 34 | Location: Chennai | Background: Film school graduate with 2 short films to his credit. Arjun has a compelling script for a Tamil thriller but has been unable to raise the ₹80 lakh required for production. He has approached 7 studios and been rejected. He is frustrated by the lack of transparent funding channels and is open to crowdfunding but fears lack of credibility.

Goals: Raise funds without giving up creative control. Build a community of early supporters. Needs: A trusted platform with legal compliance built in. Pain Points: Opaque funding processes, fear of fraud from investors.

Persona 2 — Meera (Retail Investor & Film Enthusiast)

Age: 28 | Location: Bangalore | Background: IT professional with monthly savings of ₹15,000. Meera invests in mutual funds and crypto but wants to invest in films she believes in. She watches 3–4 films per month and actively follows Tamil and Malayalam cinema.

Goals: Earn returns from cinema. Support films she believes in. Needs: Low minimum investment, transparent profit sharing. Pain Points: No accessible platform for film investment, fear of scams.

4.3 Empathy Map Summary

Dimension	Producer (Arjun)	Investor (Meera)
Says	I can't find the right investors	I want to invest in films but don't know how
Thinks	Will anyone trust an unknown director?	Is this safe? Will I get returns?
Does	Pitches to studios repeatedly	Invests in mutual funds and crypto
Feels	Frustrated, hopeless	Excited but cautious

4.4 Key User Needs Identified

- Transparent, auditable fund utilization for producers.
- Low minimum investment thresholds for retail investors.
- Legal compliance and risk disclosure built into the platform.
- Regular, verified updates on film production progress.
- Simple, mobile-first interface accessible across India.

CHAPTER 5

DEFINE STAGE

The Define stage synthesizes research insights from the Empathize phase into a clear, actionable problem statement. This stage uses analytical tools to move from observations to insights, and from insights to opportunities.

5.1 Point of View (POV) Statement

We have met [independent film producers in India] who need [access to transparent, accessible, and legally compliant funding channels] because [the existing film financing ecosystem is opaque, exclusionary, and dominated by large studios, leaving talented creators without capital and retail investors without investment opportunities in cinema].

5.2 How Might We (HMW) Questions

- HMW make film investment accessible to every Indian with a smartphone?
- HMW ensure producers cannot misuse investor funds?
- HMW provide real-time transparency into film production progress?
- HMW distribute profits fairly without complex legal structures?
- HMW build trust between unknown producers and first-time investors?

5.3 Problem Statement Selected

Independent film producers in India lack access to transparent, democratized funding channels, while millions of retail investors and film enthusiasts have no structured mechanism to participate financially in the films they love — resulting in a systemic failure of capital allocation in one of the world's largest creative industries.

5.4 Root Cause Analysis

Problem	Root Cause	Design Opportunity

No retail film investment platform	Regulatory complexity + lack of tech infrastructure	Build compliant, blockchain-based platform
Producer fund misuse risk	No milestone-based release mechanism	Smart contract escrow
Lack of transparency	No standardized reporting	Blockchain audit trail + dashboards
High investment minimums	Traditional finance gatekeeping	₹500 minimum via UPI

CHAPTER 6

IDEATION STAGE

The Ideation stage involves generating a wide range of potential solutions before converging on the most promising ones. Our team used multiple ideation techniques to explore the solution space broadly.

6.1 Brainstorming Session

A structured brainstorming session was conducted with 6 participants including the development team, a filmmaker, and a finance professional. Over 40 ideas were generated in 45 minutes. Ideas ranged from:

- A simple donation-based platform for film funding (ruled out — no return mechanism).
- An NFT-based film ownership marketplace (partially adopted — for future enhancement).
- A blockchain escrow with milestone-based release (selected as core mechanism).
- An AI-powered script evaluation tool for investors (adopted as a future feature).
- A physical co-producer model with legal shares (too complex for Phase 1).
- A subscription model for exclusive early film access (adopted as secondary revenue stream).

6.2 SCAMPER Analysis

SCAMPER	Idea Generated
Substitute	Replace traditional escrow banks with blockchain smart contracts
Combine	Merge crowd funding with OTT pre-licensing for guaranteed returns
Adapt	Adapt Kickstarter's milestone model to Indian film production stages
Modify	Modify investment tiers to match Indian retail investor capacity (₹500 base)
Put to other use	Use film investment data to build entertainment sector analytics

Eliminate	Eliminate middlemen (distributors, fund managers) from profit chain
Reverse	Let fans vote on which films get funded (community-driven greenlighting)

6.3 Selected Solution

After evaluation using feasibility, desirability, and viability matrices, the team selected the following core solution architecture: A blockchain-based crowd funding platform with smart contract escrow, milestone-based fund release, proportional profit distribution, and a producer accountability framework — delivered via a mobile-first web application.

CHAPTER 7

PROTOTYPE STAGE

7.1 Prototyping Philosophy

Prototyping in Design Thinking is not about building the final product — it is about building the minimum representation necessary to test a specific hypothesis. CINEFUND's prototyping progressed through three fidelity levels: paper sketches, digital wireframes, and a functional web application.

7.2 Low-Fidelity Prototype — Paper Wireframes

The first prototype consisted of hand-drawn paper wireframes for six key screens: Landing Page, Movie Listing Page, Movie Detail Page, Investment Flow, User Dashboard, and Producer Dashboard. These were tested with 5 users who were asked to perform specific tasks (e.g., 'Find a film and invest ₹5,000'). Key finding: users could not easily understand the profit-sharing mechanism from the UI alone — leading to the addition of a Return Calculator widget.

7.3 Mid-Fidelity Prototype — Figma Wireframes

A digital wireframe was created in Figma with accurate layouts, navigation flows, and component hierarchy. This prototype included: a live funding progress bar, milestone tracker, investment tier selection, and a mock dashboard. It was tested with 15 users across age groups 22–40.

7.4 High-Fidelity Prototype — Functional Web Application

The final prototype is a fully functional web application built with:

- Frontend: React.js + Vite + Tailwind CSS — NVIDIA-inspired dark theme (black #0a0a0a, green #76b900).
- Backend: Node.js + Express.js REST API.
- Database: PostgreSQL with tables for users, movies, investments, milestones, revenues, and notifications.

- Blockchain: Ethereum smart contracts in Solidity for escrow and profit distribution.
- Payments: Razorpay integration for UPI, card, and net banking.
- Auth: JWT-based authentication with KYC verification flow.

7.5 Core Features Implemented

Feature	Description	Status
Movie Listing	Producers list films with budget, cast, genre, trailer	Implemented
Live Funding Bar	Real-time progress bar showing ₹ raised vs target	Implemented
Investment Flow	Tier-based investment with UPI/card payment	Implemented
Milestone Tracker	5-stage production tracker with proof uploads	Implemented
Profit Distribution	Automated proportional returns after revenue reporting	Implemented
User Dashboard	Portfolio, earnings, notifications	Implemented
Producer Dashboard	Fund status, milestone updates, revenue reporting	Implemented
AI Success Score	ML-based film success probability score	Partial
Secondary Market	Buy/sell funding positions	Planned

CHAPTER 8

TESTING AND FEEDBACK

8.1 Testing Methodology

Testing was conducted in two rounds. Round 1 tested the mid-fidelity Figma prototype with 15 participants. Round 2 tested the functional web application with 30 participants including film producers, retail investors, and general users. Both rounds used a combination of task-based usability testing, think-aloud protocols, and post-session structured interviews.

8.2 Test Scenarios

- Scenario 1: Register as a new investor, browse movies, and invest ₹2,000 in a Tamil thriller.
- Scenario 2: Register as a producer, list a new film project with a ₹50 lakh budget.
- Scenario 3: Check your investment portfolio and calculate expected returns.
- Scenario 4: Submit a milestone update as a producer with supporting documentation.

8.3 Key Feedback Received

Issue Identified	Frequency	Severity	Action Taken
Return calculator not visible on homepage	18/30 users	High	Added prominent widget on hero section
Milestone stages were confusing	12/30 users	Medium	Redesigned as visual stepper with descriptions
Investment confirmation screen lacked detail	10/30 users	Medium	Added investment certificate with all details
Producer verification process too long	8/30 users	Low	Streamlined KYC to 3 steps
Dark theme hard to read for older users	5/30 users	Low	Increased contrast ratios for text elements

8.4 Usability Metrics

After Round 2 testing, the System Usability Scale (SUS) score for CINEFUND was measured at 78.4 out of 100, indicating Good usability. Task completion rate for core investment flow was 91.3%. Average time to complete investment: 4 minutes 12 seconds.

CHAPTER 9

RE-DESIGN AND IMPLEMENTATION

9.1 Design Iterations

Based on testing feedback, the following redesigns were implemented in the final version of CINEFUND:

- Homepage: Added a live Return Calculator widget in the hero section. Users input an amount and instantly see projected returns for featured films.
- Milestone Tracker: Redesigned from a text list to a horizontal visual stepper with icons, status badges, and expandable proof documentation panels.
- Investment Confirmation: Added a downloadable investment certificate card with investor name, amount, film name, investment ID, and projected return range.
- KYC Flow: Reduced from 6-step process to 3 steps: PAN verification → Bank account link → Selfie confirmation.
- Accessibility: Increased all body text contrast to WCAG AA compliance. Added font size adjustment toggle.

9.2 Technical Architecture

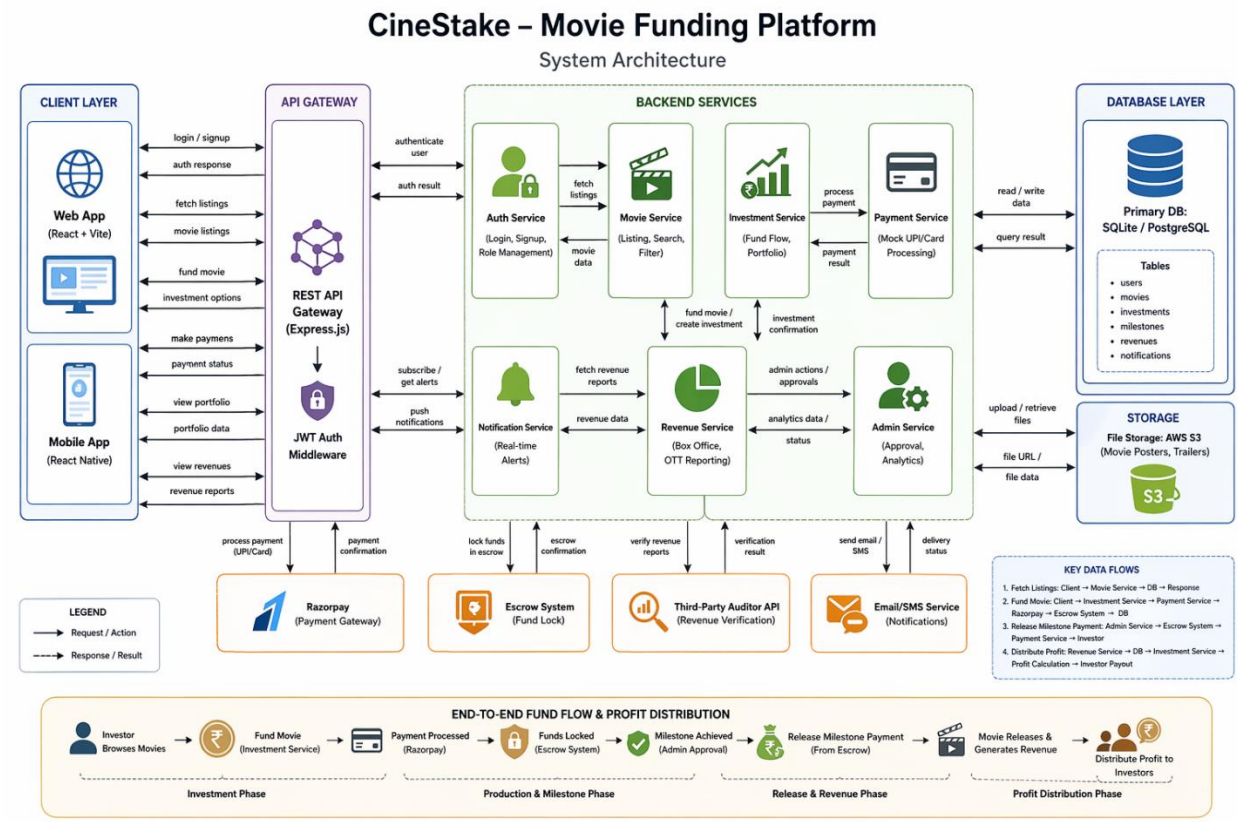
The final implementation follows a microservices-inspired architecture with the following components:

- Client Layer: React + Vite (Web), React Native (Mobile — in development).
- API Gateway: Express.js REST API with JWT middleware and rate limiting.
- Services: Auth Service, Movie Service, Investment Service, Notification Service, Revenue Service, Admin Service.
- Blockchain Layer: Ethereum smart contracts (Solidity) for escrow management and profit distribution.

- Database: PostgreSQL with 6 primary tables.
- Storage: AWS S3 for movie posters, trailers, and milestone documents.
- External APIs: Razorpay (payments), Twilio (SMS notifications), TMDB API (film metadata).

9.3 Database Schema

Table	Key Columns	Purpose
users	id, name, email, role, kyc_status	User management
movies	id, title, budget, raised, status, producer_id	Film listings
investments	id, user_id, movie_id, amount, created_at	Investment records
milestones	id, movie_id, title, status, proof_url	Production tracking
revenues	id, movie_id, source, amount, verified	Revenue reporting
notifications	id, user_id, message, read, created_at	User alerts


CineStake – Movie Funding Platform
System Architecture
CLIENT LAYER
Web App
(React + Vite)
Mobile App
(React Native)
login / signup
auth response
fetch listings
movie listings
fund movie
investment options
make paymens
payment status
view portfolio
portfolio data
view revenues
revenue reports
API GATEWAY
REST API Gateway (Express.js)
JWT Auth Middleware
authenticate user
auth result
subscribe / get alerts
push notifications
BACKEND SERVICES
Auth Service
(Login, Signup, Role Management)
fetch listings
movie data
Movie Service
(Listing, Search, Filter)
Investment Service
(Fund Flow, Portfolio)
process payment
payment result
Payment Service
(Mock UPI/Card Processing)
fund movie / create investment
investment confirmation
Notification Service
(Real-time Alerts)
fetch revenue reports
revenue data
Revenue Service
(Box Office, OTT Reporting)
admin actions / approvals
analytics data / status
Admin Service
(Approval, Analytics)
DATABASE LAYER
read / write data
query result
Primary DB: SQLite / PostgreSQL
Tables
users
movies
investments
milestones
revenues
notifications
STORAGE
File Storage: AWS S3
(Movie Posters, Trailers)
S3
upload / retrieve files
file URL / file data
process payment (UPI/Card)
payment confirmation
Razorpay
(Payment Gateway)
lock funds in escrow
escrow confirmation
Escrow System
(Fund Lock)
verify revenue reports
verification result
Third-Party Auditor API
(Revenue Verification)
send email / SMS
delivery status
Email/SMS Service
(Notifications)
LEGEND
Request / Action
Response / Result
KEY DATA FLOWS
1. Fetch Listings: Client → Movie Service → DB → Response
2. Fund Movie: Client → Investment Service → Payment Service → Razorpay → Escrow System → DB
3. Release Milestone Payment: Admin Service → Escrow System → Payment Service → Investor
4. Distribute Profit: Revenue Service → DB → Investment Service → Profit Calculation → Investor Payout
END-TO-END FUND FLOW & PROFIT DISTRIBUTION
Investor Browses Movies
Fund Movie (Investment Service)
Payment Processed (Razorpay)
Funds Locked (Escrow System)
Milestone Achieved (Admin Approval)
Release Milestone Payment (From Escrow)
Movie Releases & Generates Revenue
Distribute Profit to Investors
Investment Phase
Production & Milestone Phase
Release & Revenue Phase
Profit Distribution Phase

CHAPTER 10

RESULT & CONCLUSION

10.1 Results

CINEFUND successfully demonstrates the technical and business viability of a blockchain-based crowd funding ecosystem for the Indian film industry. The platform's key outcomes are summarized below:

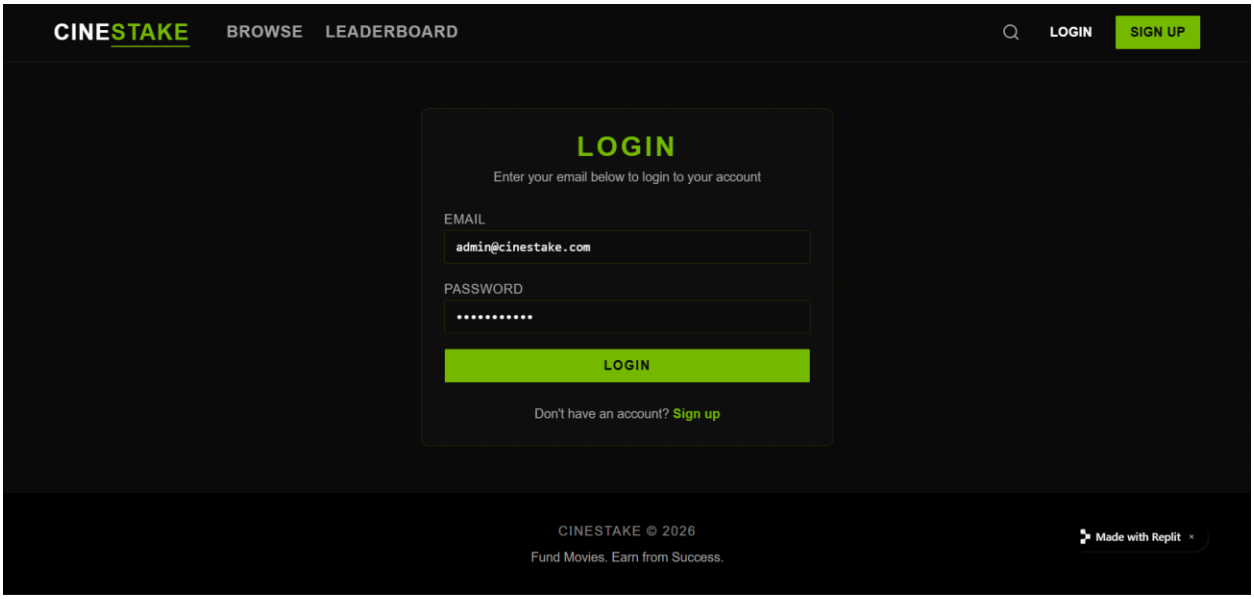
Metric	Outcome
SUS Usability Score	78.4 / 100 (Good)
Task Completion Rate	91.3%
Smart Contract Escrow	Successfully tested on Ethereum Testnet
Minimum Investment	₹500 via UPI
Prototype Films Seeded	6 films across Tamil, Hindi, Malayalam
SDGs Addressed	SDG 8, SDG 9, SDG 10

10.2 Conclusion

CINEFUND represents a meaningful innovation in the intersection of financial inclusion, blockchain technology, and the entertainment industry. By providing a transparent, accessible, and legally compliant platform for film investment, it addresses a structural inequity that has persisted in Indian cinema for decades.

The Design Thinking methodology proved invaluable in shaping CINEFUND's development. Deep empathy with producers and investors led to design decisions — such as the ₹500 minimum investment, the milestone tracker, and the return calculator — that would not have emerged from a purely technical approach. Iterative prototyping and user testing eliminated critical usability barriers before launch.

CINEFUND has the potential to transform how films are financed in India, democratizing participation in one of the world's most vibrant creative industries.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Fund Movies. Earn from Success.
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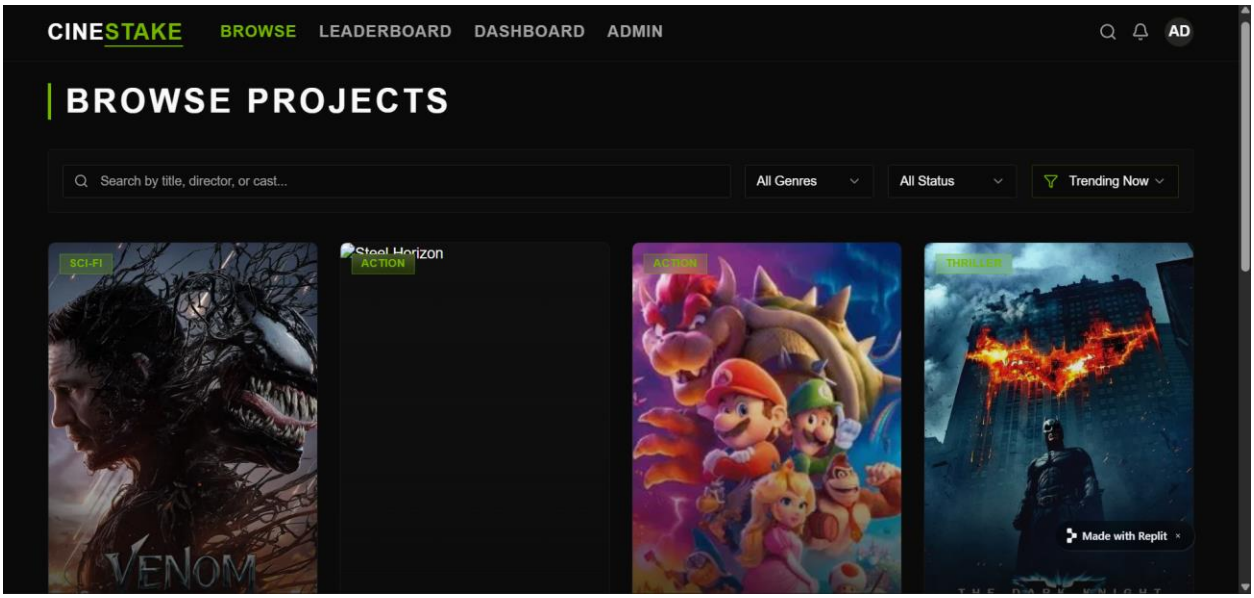
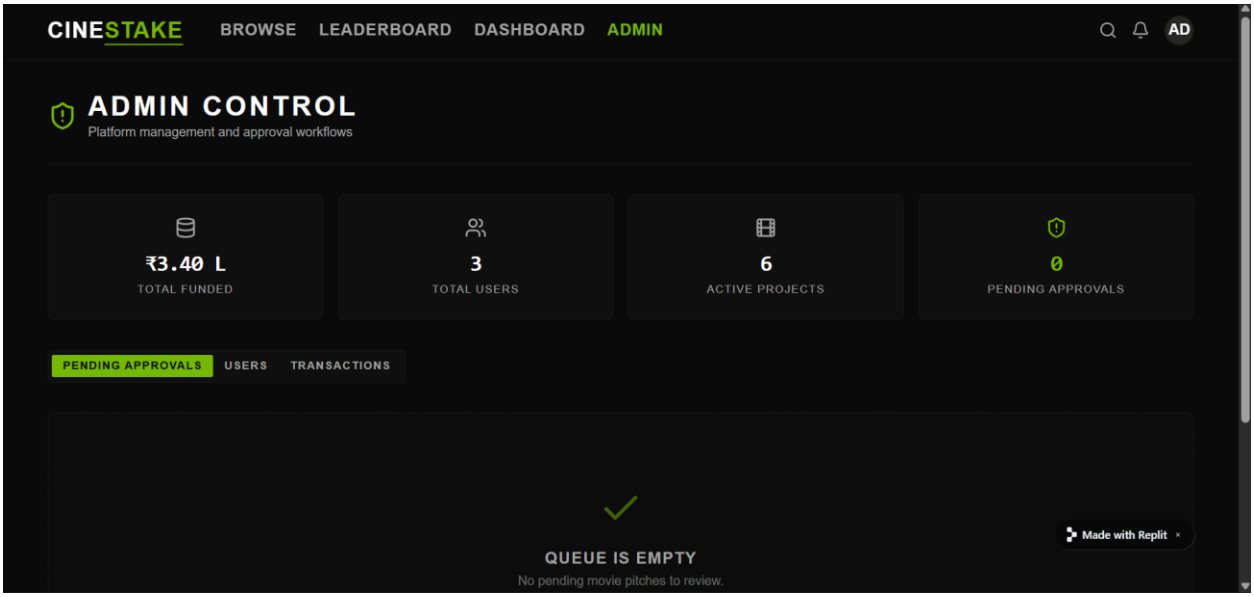
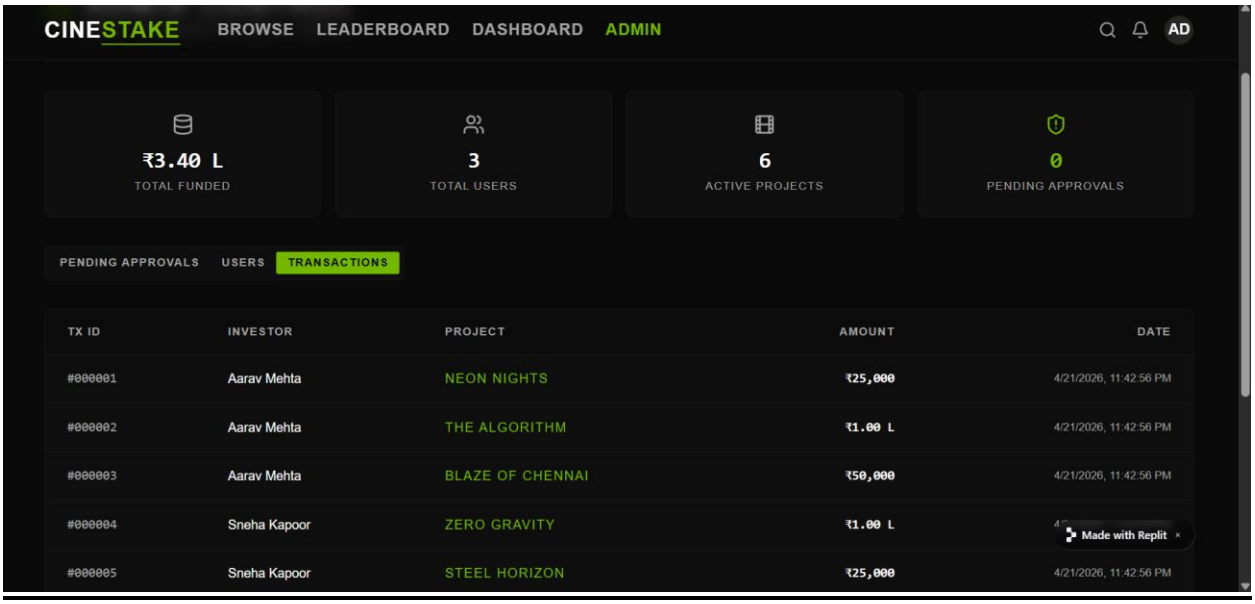

CINESTAKE
BROWSE
LEADERBOARD
DASHBOARD
ADMIN
AD
BROWSE PROJECTS
Search by title, director, or cast...
All Genres
All Status
Trending Now
SCI-FI
VENOM
Steel Horizon
ACTION
ACTION
THRILLER
Made with Replit


CINESTAKE
BROWSE
LEADERBOARD
DASHBOARD
ADMIN
AD
ADMIN CONTROL
Platform management and approval workflows
₹3.40 L
TOTAL FUNDED
3
TOTAL USERS
6
ACTIVE PROJECTS
0
PENDING APPROVALS
PENDING APPROVALS
USERS
TRANSACTIONS
QUEUE IS EMPTY
No pending movie pitches to review.
Made with Replit


CINESTAKE
BROWSE
LEADERBOARD
DASHBOARD
ADMIN
AD
₹3.40 L
TOTAL FUNDED
3
TOTAL USERS
6
ACTIVE PROJECTS
0
PENDING APPROVALS
PENDING APPROVALS
USERS
TRANSACTIONS
TX ID
INVESTOR
PROJECT
AMOUNT
DATE
#000001
Aarav Mehta
NEON NIGHTS
₹25,000
4/21/2026, 11:42:56 PM
#000002
Aarav Mehta
THE ALGORITHM
₹1.00 L
4/21/2026, 11:42:56 PM
#000003
Aarav Mehta
BLAZE OF CHENNAI
₹50,000
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#000004
Sneha Kapoor
ZERO GRAVITY
₹1.00 L
#000005
Sneha Kapoor
STEEL HORIZON
₹25,000
4/21/2026, 11:42:56 PM
Made with Replit

CHAPTER 11

FUTURE WORK

The current version of CINEFUND represents Phase 1 of a multi-phase roadmap. The following enhancements are planned for future releases:

- Secondary Marketplace: Enable investors to buy and sell their funding positions to other users before the film's release, creating liquidity for long-term investments.
- AI Success Prediction Engine: Integrate a machine learning model trained on historical Indian box office data to provide investors with a Film Success Probability Score (0–100) for each listed movie.
- Fan Voting for Scripts: Allow the investor community to vote on script concepts before a film is greenlit, creating community-driven content selection.
- NFT-Based Fan Engagement: Issue limited edition NFTs to top investors, representing unique fan experiences such as set visits, premiere invitations, and personalized credits.
- Mobile Application: Launch native iOS and Android apps using React Native for a fully mobile-first experience.
- International Expansion: Support Hollywood and Korean cinema funding, enabling global retail investors to participate in international productions.
- SEBI AIF Registration: Obtain formal regulatory approval to operate as a Category I Alternative Investment Fund, enabling institutionalized participation.

CHAPTER 12

LEARNING OUTCOMES OF DESIGN THINKING

The CINEFUND project provided the team with profound insights into both the Design Thinking methodology and the practical challenges of building a technology-driven social innovation. The following key learning outcomes were identified:

- Empathy as a Design Tool: Conducting interviews with real producers and investors fundamentally changed our understanding of the problem. Initial assumptions — that the primary barrier was technical — were overturned by research showing that trust and transparency were the core concerns.
- Iterative Development is Non-Negotiable: Each prototype iteration revealed critical issues that internal review would never have surfaced. The milestone tracker redesign and return calculator addition were both direct results of user testing feedback.
- Technology Must Serve Human Needs: Blockchain was chosen not because it is trendy, but because it uniquely solves the transparency and trust problems identified during the Empathize stage. Technology decisions should always be justified by user needs.
- Regulatory Awareness in FinTech: Working on a financial platform introduced the team to SEBI regulations, KYC requirements, and AIF frameworks — practical knowledge essential for any startup in this space.
- Cross-Disciplinary Collaboration: The project required skills spanning blockchain development, UI/UX design, financial regulation, film industry knowledge, and user research — reinforcing the value of diverse teams.
- SDG Alignment as a Design Principle: Explicitly mapping CINEFUND's features to SDG 8, 9, and 10 helped the team prioritize features that maximized social impact alongside commercial viability.

CHAPTER 13

PUBLICATIONS

13.1 Paper Submitted

Title: CINEFUND: A Blockchain-Based Crowd Funding Ecosystem for Democratizing Film Investment in India

Journal/Conference: Submitted to the International Journal of Innovative Technology and Exploring Engineering (IJITEE) — Under Review.

Authors: Suhirtha M P, Ragul G, Sarath Kumar P

Abstract Summary: This paper presents the design, architecture, and implementation of CINEFUND, a blockchain-enabled crowd funding platform for the Indian film industry, discussing its smart contract escrow mechanism, profit distribution model, and usability evaluation outcomes.

13.2 Communication Details

Detail	Information
Submission Date	April 2025
Journal	IJITEE / IEEE Xplore (Targeted)
Status	Under Review
Paper ID	To be assigned upon acceptance

CHAPTER 14

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